Toward a Globally-Applicable Uncertainty Quantification Framework for Satellite Multisensor Precipitation Products Based on GPM DPR

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***Abstract*— The usefulness of satellite multisensor precipitation products such as NASA's 30-min, 0.1° Integrated Multi-satellitE Retrievals for the Global Precipitation Measurement (GPM) mission (IMERG) is hindered by their associated errors. Reliable estimates of uncertainty would mitigate this limitation, especially in near-real time when gauge observations are not available. However, creating such estimates is challenging, due to both the complicated nature of satellite precipitation errors and the lack of "ground-truth" data precisely in the places—including oceans, complex terrain, and developing countries—that could benefit most from satellite precipitation estimates. In this work, we use the GPM dual-frequency precipitation radar (DPR)-derived swath-based precipitation products as an alternative to ground-based observations to facilitate IMERG uncertainty estimation. We compare the suitability of two DPR-derived precipitation products, 2ADPR and 2BCMB, against higher fidelity ground validation multiradar multisensor (GV-MRMS) ground reference data over the contiguous United States. The 2BCMB is selected to train error models based on censored shifted gamma distribution (CSGD; a mixed discrete-continuous probability distribution). Uncertainty estimates from these models are compared against alternative error models trained on GV-MRMS. Using information from NASA's Modern-Era Retrospective analysis for Research and Applications, version 2 (MERRA-2) reanalysis, we also demonstrate how IMERG uncertainty estimates can be further constrained using additional precipitation-related predictors. Though several critical issues remain unresolved, the proposed method shows promise for yielding robust uncertainty estimates in near-real time for IMERG and other similar precipitation products at their native resolution across the entire globe.**

***Index Terms*— Dual-frequency precipitation radar (DPR), error model, Global Precipitation Measurement (GPM) mission, satellite multisensor precipitation (SMP), uncertainty.**

I. Introduction

THE potential of using satellite precipitation estimates to understand and inform predictions of global-to-regional water cycles has been recognized for decades [1], [2], [3]. Due to the limited number and uneven distribution of rain gauges that accurately measure precipitation on the ground [4], global satellite multisensor precipitation (SMP) products have been increasingly applied to support decision-making, particularly in data-sparse regions such as the oceans, mountainous areas, and developing countries (see [5], [6], [7]).

SMP products generally merge measurements from passive microwave (PMW) and infrared (IR) sensors to create consistent high-resolution gridded precipitation estimates [8], [9]. A number of global SMP products have been developed based on different merging methods, including the climate prediction center morphing technique (CMORPH) [10], [11], the Integrated MultisatellitE Retrievals for the Global Precipitation Measurement (GPM) mission (IMERG) [12], and its predecessor—the Tropical Rainfall Measuring Mission (TRMM) Multisatellite Precipitation Analysis (TMPA) [13] and the precipitation estimation from remotely sensed information using artificial neural networks (PERSIANN) family [14], [15].

Despite continual improvements, the usefulness of SMP products remains limited due to their oftentimes poor accuracy at fine space and time scales [16], [17]. These errors stem from a variety of sources, including heterogeneous sensor

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properties [18], [19], retrieval algorithm deficiencies [20], and insufficient spatial and temporal sampling [21], [22]. The absolute and relative roles of these error sources can depend on season, precipitation intensity, storm type, geophysical features such as altitude and land surface type, and other factors [23], [24], [25].

A large number of existing studies have presented empirical characterizations of SMP errors using "ground truth," i.e., more reliable reference observations (typically rain gauges or gauge-corrected weather radar; see [17] for a discussion). Errors are often separated into systematic (i.e., bias) and random error components [26], [27], [28]. They typically depend on precipitation magnitude via conditional bias (heteroscedasticity) in the case of systematic (random) error [17]. Other approaches have considered additional terms to characterize errors in both detection and magnitude estimation, distinguishing between "false alarm" precipitation, missed precipitation, and hit bias [29]. These studies have always been undertaken at local to regional scales due to the lack of sufficient ground reference globally [30], [31], [32], [33]. Unfortunately, however, lessons learned in such studies cannot be easily transferred to other places due to the complexity of satellite precipitation uncertainties [22], [34].

Furthermore, ex-post SMP error studies are not sufficient to meet uncertainty characterization requirements for applications, particularly those in near-real time. Such a requirement has recently been prioritized by the IMERG development team—specifically, to provide uncertainty estimates at IMERG's native 30-min, 0.1° resolution that can be included in IMERG data files at the time of creating these files [12], [35].

A more limited number of studies have sought to develop so-called error models that attempt to characterize the uncertainty associated with any particular SMP product, generally expressed in the form of a probability distribution of "true precipitation" (see [36], [37]). Error model development is challenging due in part to the mixed discrete-continuous nature of intermittent precipitation, an issue that becomes increasingly important to address as SMP products advance to higher spatial and temporal resolutions. Some error models just ignore intermittency altogether to focus on hit biases and random errors (see [28], [36]), while others have attempted to address it (see [38], [39], [40]), but arguably at the expense of relatively complicated formulations and limited flexibility [37]. An alternative approach has also been proposed to characterize uncertainty as an integral part of SMP retrieval algorithms and to subsequently yield probabilistic precipitation estimates [25].

Regardless of the specific model formulation, the availability of ground reference data to train these error models has posed a fundamental limitation, since reference measurements are lacking precisely in the locations (i.e., data-sparse regions) that could benefit most from spaceborne remote sensing [24]. It is, thus, highly desirable to explore globally applicable uncertainty quantification approaches that can perform anywhere, even in the total absence of local or regional ground reference observations.

In this study, we explore the idea that the dual-frequency precipitation radar (DPR) on board the GPM Core Observatory (GPM-CO) can be utilized in place of ground reference data. If valid, this facilitates the development of worldwide native-resolution error estimates of IMERG as well as other similar SMP products. Recent studies have explored the potential for DPR as an alternative reference to evaluate PMW-only precipitation estimates [41], [42] and merged precipitation products [43]. This study advances that concept to propose a prototype uncertainty quantification framework for IMERG. We use two DPR-derived precipitation products and co-located IMERG estimates to train a parsimonious mixed discrete-continuous error model. These DPR-derived products trained error models are evaluated against alternative models trained on ground reference observations over the contiguous United States (CONUS). The error model can also incorporate additional predictors. We examine whether a global reanalysis dataset, the Modern-Era Retrospective analysis for Research and Applications, version 2 (MERRA-2), can further constrain IMERG uncertainties. As far as we are aware, this is the first study to explore the feasibility of a globally applicable prototype framework for uncertainty quantification at the IMERG native resolution, though we leave global validation and several other important details to future work.

The datasets used in this study are described in Section II. The data resampling and matching algorithm, the error model, and evaluation metrics are introduced in Section III. Section IV presents the results; discussion follows in Section V. A summary and conclusions are provided in Section VI.

II. Data

We selected CONUS as the study area (Fig. 1) for two reasons: first, it is covered by a high-quality, high-resolution ground reference precipitation product that allows us to validate the proposed approach; second, its large geographic extent and climatic diversity allow a relatively comprehensive assessment of the approach's robustness. The study period is June 2014 to April 2019 (∼five years). No attempt is made to address seasonally varying uncertainty, nor to discriminate by precipitation phase. Prior studies have argued that the former may not be critical [37], [40], while the latter certainly is.

A. IMERG

IMERG unifies all PMW observations from the GPM constellation with augmentation from IR data to produce 30-min, 0.1° gridded precipitation estimates over the entire globe [19], [44]. The sampling gaps between infrequent PMW overpasses are filled by a time-weighted "morphing" interpolator using propagation vectors calculated from ancillary data (previously IR data, and switch to numerical model simulated total column precipitable water vapor since version 06; [45]). This morphing technique was first developed as part of CMORPH, including forward and backward propagation of PMW scans in time [10]. PMW-only precipitation estimates are generated using the Goddard profiling algorithm (GPROF) [46], and then morphed and supplemented, via a Kalman filter [11], with PMW-calibrated IR precipitation estimates from PERSIANN algorithm [47]. Three IMERG runs—Early

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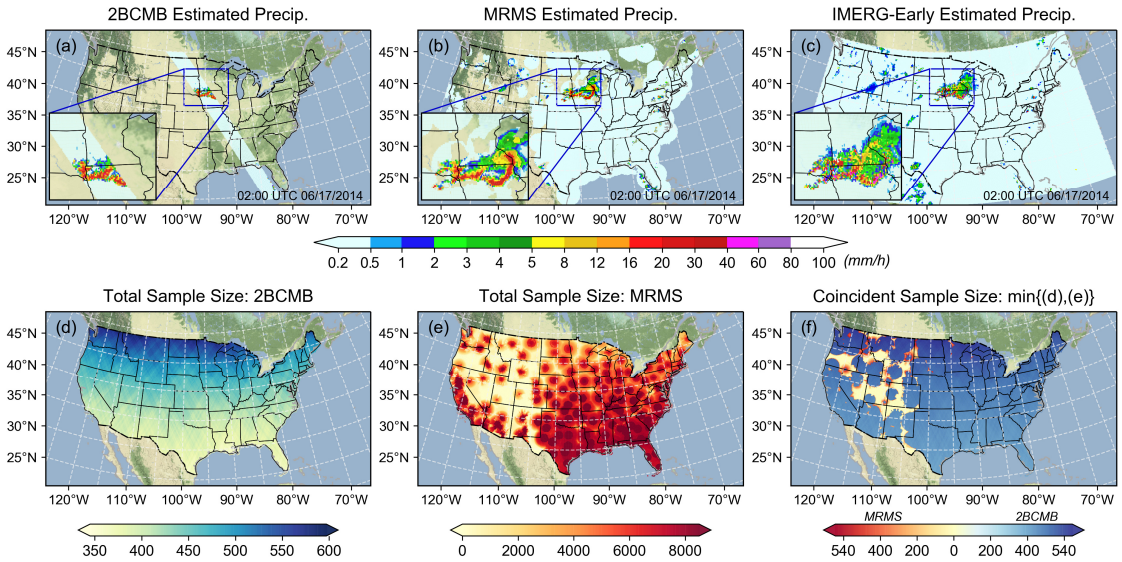

Fig. 1. Coincident precipitation estimates from regridded: (a) 2BCMB (2ADPR is similar; see Supplemental Fig. S1); (b) GV-MRMS; and (c) IMERG-E for 02:00–02:30 UTC June 17, 2014, with the maps for the total sample size of: (d) 2BCMB (also 2ADPR); (e) GV-MRMS; and (f) coincident sample size—the minimum from (d) and (e)—within 0.1° × 0.1° boxes during the study period.

(hereafter IMERG-E), Late, and Final—address different user requirements for latency and accuracy, and this study focuses on version 06B IMERG-E. IMERG-E excludes the backward propagation in the morphing, and thus, the waiting time for extra PMW overpasses. For this reason, it is arguably the most useful for near-real-time applications due to its short latency (4 h, compared to 12 h and three months for Late and Final, respectively) but features the largest errors due to the more limited use of PMW information.

While the IMERG processing algorithm consists of many elements beyond the scope of this study, it is worth mentioning that it uses observations from the DPR and GPM Microwave Imager (GMI) on board the GPM-CO. Microwave radiances from all partner constellation PMW sensors are calibrated to GMI for a bias-corrected, consistent radiometric dataset before retrieving precipitation rates [48]. Then, the combined DPR and GMI data product from the GPM COmbined Radar–Radiometer Algorithm (CORRA) [49] contributes to IMERG in terms of its derived hydrometeor profiles and surface precipitation. These two elements are used to construct a priori databases in GPROF to convert the calibrated PMW radiances into precipitation, while the latter is further used to calibrate the PMW-only precipitation estimates on a rolling 45-day basis over ocean (the calibration over land is based on the Global Precipitation Climatology Project data; see [12]). We mention this because it constitutes a potential objection to the usage of DPR (and, as the reader will see, GMI) as the reference for uncertainty estimation due to a possible lack of independence between IMERG and those instruments. This issue is discussed further in Sections IV-A and V-A.

B. Ground Reference: GV-MRMS

The ground validation multiradar/multisensor (GV-MRMS; [50], [51]) dataset is derived from the MRMS system that combines the polarimetric WSR-88D CONUS radar network with rain gauges and other auxiliary information to generate high-resolution quantitative precipitation estimates (QPE) over CONUS [52]. GV-MRMS QPE has been used as a ground reference for evaluation of various satellite precipitation products [20], [53], [54], [55]. In this study, we use the Level-3 regridded GV-MRMS QPE product, which was created specifically to support GPM ground validation [20]. This QPE includes a 30-min, 0.01° gauge-corrected precipitation rate (GCP) as well as a radar quality index (RQI) which ranges from 0 to 100, with 100 representing the best quality.

C. DPR-Based Reference Precipitation Datasets

We consider two recent (version 06) GPM Level-2 DPR products as potential alternatives to a ground-based reference: 2ADPR and 2BCMB. Both provide high-resolution (approximately 5-km DPR footprint diameter at nadir) precipitation estimates on an instantaneous field of view between 65° N and 65° S. The 2ADPR is derived based on Ku-band (13.6 GHz) and Ka-band (35.5 GHz) DPR measurements, and it uses dual-frequency observations to infer precipitation phase and reconstruct 3-D hydrometeor and precipitation fields [56], [57]. This study uses the 2ADPR data field "precipRateESurface," which is extrapolated from the lowest clutter-free DPR bin to estimate surface precipitation rate [58]. The 2BCMB, on the other hand, combines DPR reflectivities and GMI radiances using the CORRA algorithm to offer high-quality precipitation estimates from GPM [48]. We use the 2BCMB data field "surfPrecipTotRate" in the following analysis. It is worth noting that these DPR-derived reference datasets are distinct from the DPR and GMI combined data that have been used in GPROF (for version 06B IMERG; see Section II-A) since the latter draws from version 04 of 2BCMB [59].

Both 2ADPR and 2BCMB data fields are obtained from normal scans (i.e., the widest swath scans from DPR; [57]) to maximize sample size. While 2BCMB and 2ADPR present

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different error structures [60], postlaunch evaluations showed that DPR and GMI can detect precipitation rates down to 0.1 mm·h^{-1} (see [41], [61]). This precipitation rate was, thus, selected as the rain/no-rain detection threshold in this study for all datasets.

D. Additional Predictors: MERRA-2

We also examine the potential to further constrain the uncertainty estimates by incorporating additional predictors. Following previous findings [37], [62], [63], [64], in this study, we consider the total precipitable water vapor (TQV), topmost soil layer's ground wetness index ($GWET_{TOP}$), and 2-m air temperature (T2M) from NASA's MERRA-2 reanalysis product [65]. To match with the above datasets, this study uses 0.5° (latitude) × 0.625° (longitude), hourly MERRA-2 outputs. $GWET_{TOP}$ is a dimensionless relative saturation index for the upper 5 cm of soil. Based on previous studies showing that soil moisture changes can enhance satellite precipitation estimation [63], [64], we also derive a variable we call $GWETD_{TOP}$, which is the difference between the current and preceding hour's values of $GWET_{TOP}$. Negative values of $GWETD_{TOP}$ correspond to soil evaporation, while positive values indicate precipitation occurrence. We transform all the negative $GWETD_{TOP}$ values to zero before including it as a predictor in the uncertainty framework.

It should be emphasized here that our goal was not to identify the best possible additional predictors, but rather to simply illustrate that such predictors could be utilized to constrain IMERG uncertainty estimates. This issue is discussed further in Section V-C.

III. Methodology

A. Matching and Preprocessing of Multiple Datasets

Following the approach of Khan et al. [43], GV-MRMS, 2ADPR, 2CBMB, and MERRA-2 data are matched into the IMERG 0.1° × 0.1° grid (∼10 km) on the IMERG half-hourly interval. GV-MRMS is upscaled by averaging all grid cells in a 10 × 10 window, provided that the RQI for at least 90% of these pixels is 100. This guarantees reliable ground reference, but it also inevitably biases the coincident data samples toward certain locations and synoptic environments due to the dependence of RQI on these conditions [66]. The two DPR-derived precipitation products are regridded by averaging all the DPR footprint scale (∼5 km) estimates falling within the IMERG grid cell while MERRA-2 is mapped into it using nearest neighbor interpolation. For temporal alignment, the two DPR-derived precipitation products are matched into the appropriate 30-min IMERG observation interval, and hourly MERRA-2 data are duplicated for the two half hours. Fig. 1 (top panels) shows an example of the regridded coincident precipitation estimates from 2BCMB, GV-MRMS, and IMERG-E.

The sample size of DPR-based precipitation products generally decreases from North to South due to the inclined orbit of GPM [Fig. 1(d)], while GV-MRMS data are limited in western CONUS because of radar beam blockage, and more generally by distance between radars in flatter terrain [Fig. 1(e)]. The coincident data sample size, thus, depends on location and is generally less than 600 in each 0.1° grid cell [Fig. 1(f)]. To ensure a sufficiently large sample size, error models are trained and validated by pooling all coincident 0.1° data samples within 1° × 1° spatial windows. In addition, the number of precipitating (i.e., nonzero) samples affects the stability of error modeling, particularly at dry locations (e.g., western CONUS). The majority of precipitation observations are zero at those places and thus carry only very limited information that can be leveraged to train the error models [67]. In this regard, for windows of western CONUS where the sample size is less than 5000, we further pool data from the adjacent four windows in the East–West and North–South directions for model fitting.

B. CSGD-Based Uncertainty Quantification Framework

The uncertainty quantification framework selected in this study follows the censored shifted gamma distribution (CSGD) method developed by Scheuerer and Hamill [67] for post-processing ensemble numerical precipitation forecasts. It was adapted by Wright et al. [37] to characterize the uncertainty for daily-scale satellite precipitation estimates. The CSGD can simultaneously depict precipitation occurrence and magnitude by introducing a "shift" parameter δ ($\delta < 0$) into the conventional two-parameter gamma distribution $F_{\mu,\sigma}$ (here, it is parameterized by mean μ and standard deviation σ, rather than shape and scale/rate parameters). The cumulative distribution function (CDF) of the CSGD is left-censored at zero

$$F_{\mu,\sigma,\delta}(x) = \begin{cases} F_{\mu,\sigma}(x-\delta), & \text{for } x \geq 0 \\ 0, & \text{for } x < 0 \end{cases} \tag{1}$$

where x is the precipitation rate (mm·h^{-1}). The vertical intercept $F_{\mu,\sigma,\delta}(0)$ is one minus the probability of precipitation (POP), and the CDF to the right of zero represents the nonexceedance probabilities associated with nonzero precipitation rates.

The CSGD-based error model consists of two main pieces: first, a "climatological CSGD" with parameters μ, σ, and δ in (1) and second, a regression system that comprises the error model. Once trained, this regression system can produce an estimated "conditional" CSGD with parameters $\mu(t)$, $\sigma(t)$, and $\delta(t)$ [see (2)–(4)] that represents the possible true precipitation rate and POP conditioned on an IMERG-E retrieval (and other optional predictors) at time t. This regression system can capture both conditional bias and heteroscedasticity, as well as the discrete-continuous nature of precipitation and associated errors. The most basic regression system lets $\mu(t)$ increase linearly with IMERG-E magnitude $P_I(t)$, and all error models used here assume that $\sigma(t)$ is proportional to the square root of $\mu(t)$ (see [67]). We will refer to this most basic variant as the "linear error model"

$$\mu(t) = \mu\left[\alpha_2 + \alpha_3 \frac{P_I(t)}{\bar{P}_I}\right] \tag{2}$$

$$\sigma(t) = \alpha_4 \sigma \sqrt{\frac{\mu(t)}{\mu}} \tag{3}$$

$$\delta(t) = \delta \tag{4}$$

where $\bar{P}_I$ denotes the climatological IMERG-E mean.

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The linearity assumption can be relaxed to account for nonlinear conditional bias. This version (hereafter "nonlinear error model") replaces (2) with

$$\mu(t) = \frac{\mu}{\alpha_1} \log 1\mathrm{p}\left\{ \exp\ \mathrm{m}1(\alpha_1)\left[\alpha_2 + \alpha_3 \frac{P_1(t)}{\bar{P}_1}\right]\right\} \tag{5}$$

where $\log 1\mathrm{p}(x) = \log(1 + x)$ and $\exp\mathrm{m}1(x) = \exp(x) - 1$.

Both the linear and nonlinear models can also accommodate extra time-varying predictors or covariates $C(t)$, potentially further constraining (i.e., narrowing) uncertainty estimates. To this end, (2) and (5) can be replaced with

$$\mu(t) = \mu\left[\alpha_2 + \alpha_3 \frac{P_1(t)}{\bar{P}_1} + \alpha_5 \frac{C(t)}{\bar{C}}\right] \tag{6}$$

$$\mu(t) = \frac{\mu}{\alpha_1} \log 1\mathrm{p}\left\{ \exp\ \mathrm{m}1(\alpha_1)\left[\alpha_2 + \alpha_3 \frac{P_1(t)}{\bar{P}_1} + \alpha_5 \frac{C(t)}{\bar{C}}\right]\right\} \tag{7}$$

respectively, where $\bar{C}$ is the climatological mean of the covariate. While several covariates can be used simultaneously (not depicted in (6) and (7); [37], [67]), this study only considers covariates individually.

All of the above regression coefficients (α_1–α_5) as well as the three CSGD parameters are optimally estimated using the techniques detailed in [67], which minimize the continuous ranked probability score (CRPS) between empirical and theoretical CDFs.

C. Error Model Training and Validation

The 2ADPR and 2BCMB are first compared against coincident GV-MRMS observations over CONUS. This comparison considers the ability to correctly detect precipitation occurrence and to estimate precipitation rates of hit cases. To evaluate precipitation occurrence, we create contingency tables [68] showing the numbers and rates of hits, misses, false alarms, and correct nondetects (CNs) using 0.1 mm·h^{-1} as the detection threshold (see Section II-C). Precipitation rates for hits are then assessed for every 1° × 1° spatial window in terms of three evaluation metrics: the relative bias (RB), root-mean-squared error (RMSE), and Pearson's correlation coefficient (CC), which have been widely used in previous studies (see [43], [69]).

The regridded coincident datasets are randomly divided with 80% of observations used for CSGD-based error model training and 20% for validating model performance. A range of error model complexities are explored: linear models [i.e., (1)–(4)], nonlinear models [i.e., (1) and (3)–(5)], linear models with a single covariate [i.e., (1), (3), (4), and (6)], and nonlinear models with a single covariate [i.e., (1), (3), (4), and (7)]. All the error models are trained using both the selected DPR-derived precipitation data (2ADPR or 2BCMB) and GV-MRMS, while the latter are used as performance benchmarks to evaluate if the former trained error model appears reasonable.

DPR-derived products and GV-MRMS trained error models are then applied to the validation dataset, and their conditional CSGD estimates of reference precipitation are evaluated against GV-MRMS observations from that dataset using both deterministic and probabilistic metrics. To examine if the error model can effectively characterize the central tendency of IMERG-E errors, we compare the conditional CSGD median with GV-MRMS using mean absolute error (MAE)

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - o_t| \tag{8}$$

where y_t is either the CSGD median or IMERG-E, o_t is the coincident GV-MRMS observation at time t, and n is the number of (y_t, o_t) pairs.

Similar to other deterministic evaluation metrics (e.g., RB, RMSE, and CC), MAE is insufficient for fully characterizing the predicted probabilistic distributions from CSGD error models. CRPS, on the other hand, measures the dispersion of these distributions around a GV-MRMS observation. CRPS, thus, offers a probabilistic performance measure of the error models

$$\mathrm{CRPS}(F_{\mu(t),\sigma(t),\delta(t)}, o_t) = \int_{-\infty}^{\infty} [F_{\mu(t),\sigma(t),\delta(t)}(x) - I(o_t \le x)]^2 dx \tag{9}$$

where $F_{\mu(t),\sigma(t),\delta(t)}$ denotes the CDF of the conditional CSGD model at time t, and $I(\cdot)$ is a step function that takes the value of 1 if $x \ge o_t$ (i.e., GV-MRMS observation at time t) and 0 elsewhere. Low CRPS indicates that the predicted CSGD's density is concentrated relatively close to the reference, while high CRPS implies either a very "wide" distribution or one that is heavily biased. Note that CRPS is mathematically identical to MAE for deterministic—as opposed to probabilistic—predictions.

Heteroscedasticity in IMERG-E errors means that we should not simply compare or combine CRPS values across various locations, since, like MAE or RMSE, CRPS will tend to be larger for heavier precipitation regimes. This has three implications for our error model's validation. First of all, we apply "reduction CRPS" (RCRPS) [70] for comparing model performance across different locations (i.e., 1° × 1° boxes). It is normalized by the standard deviation of GV-MRMS observations at that location (denoted as σ_{M}) and thus is dimensionless

$$\mathrm{RCRPS} = \frac{\mathrm{CRPS}}{\sigma_M}. \tag{10}$$

Second, the validation dataset is then grouped into four categories: hits, misses, false alarms, and CNs, by comparing the coincident IMERG-E and GV-MRMS data (using the same 0.1 mm·h^{-1} threshold). CRPS is then calculated for each group to evaluate model performance under different cases. In addition, the calculated CRPSs of hit cases are further grouped by the magnitude of IMERG-E, to investigate the magnitude-dependent performance of the uncertainty estimates from different error models.

Finally, we further examine the performance of the error models' probabilistic estimates of precipitation events given a number of thresholds using reliability diagrams (see [68] for details). Considering GV-MRMS observation o_t and the CDF of CSGD error model $F_{\mu(t),\sigma(t),\delta(t)}$ at time t, the observed and

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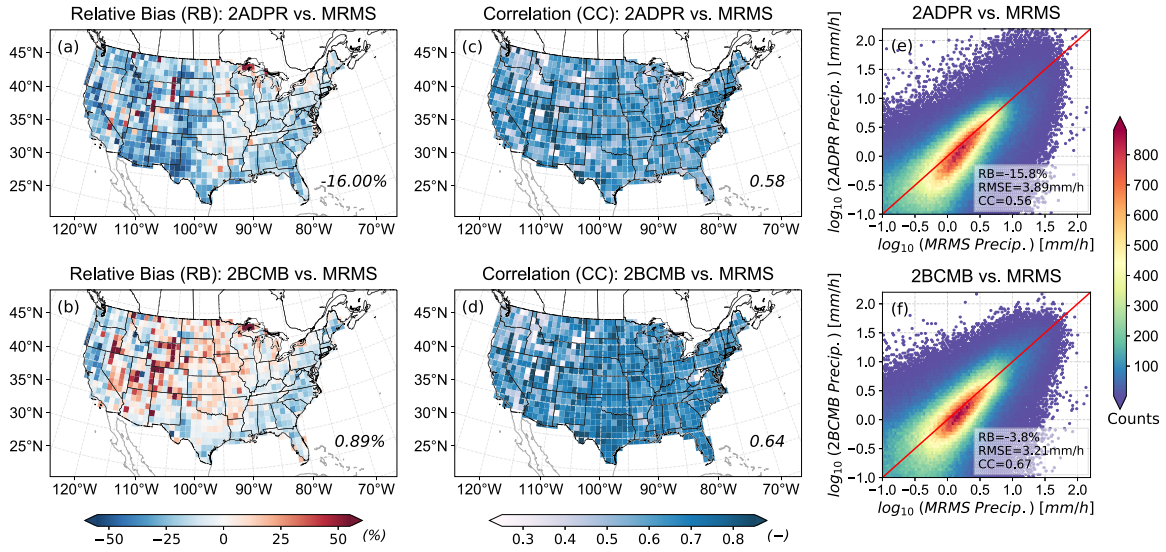

Fig. 2. Spatial maps of: (a) and (b) RB; (c) and (d) CC; and (e) and (f) density scatterplots by comparing the coincident precipitation estimates from 2ADPR and 2BCMB versus GV-MRMS during the study period. Only precipitation estimates greater than 0.1 mm·h^{-1} are considered. Inset values in (a)–(d) are the mean across all grid boxes (1° × 1°) over CONUS.

predicted probability that an "event" occurred can be defined

$$X_{e,t} = \begin{cases} 1, & \text{for } o_t > \text{TH} \\ 0, & \text{for } o_t \le \text{TH} \end{cases} \tag{11}$$

$$Y_{e,t} = 1 - F_{\mu(t),\sigma(t),\delta(t)}(\text{TH}) \tag{12}$$

where TH is the threshold (mm·h^{-1}). $X_{e,t}$ denotes the observed probability of threshold exceedance (either 0 or 1), while $Y_{e,t}$ is the predicted probability of threshold exceedance (between 0 and 1) from the error model. Following previous studies (see [71], [72]), we sort and group all predicted probabilities $Y_{e,t}$ (e.g., $t = 1, 2, \ldots, N$ for validation dataset) into ten equally-sized bins (0–10%, 10%–20%, . . . , 90%–100%). For each group, both the average predicted probability and the average observed probability are calculated. In a reliability diagram, a perfect prediction model would yield results that fall along the 1:1 line. For example, when $Y_{e,t} = 0.90$, we expect the event to occur 90% of the time in reality. All coincident samples across CONUS are pooled for this analysis.

IV. Results

A. DPR-Derived Precipitation Products as Reference

Fig. 2 shows CONUS-wide "hits only" evaluation results of DPR-derived precipitation products against coincident GV-MRMS observations. The 2ADPR underestimates precipitation almost everywhere, particularly in the western parts of the country leading to a CONUS-wide average underestimation of 16% [Fig. 2(a)]. The 2BCMB, on the other hand, varies geographically with overestimation (e.g., the Rockies and Great Plains) and modest underestimation (e.g., the West and East Coasts) leading to a CONUS-wide average within 1% of GV-MRMS [Fig. 2(b)]. Moreover, 2BCMB is better correlated with GV-MRMS observations over most of CONUS [Fig. 2(c) and (d)]. Scatterplots and three summary statistics (RB, RMSE, and CC) again indicate that 2BCMB generally outperforms 2ADPR [Fig. 2(e) and (f)]. The 2ADPR, however, shows somewhat better detection skills—lower numbers of false alarms and missed precipitation (Table I). From Table I, however, it can be seen that hits are more common than false alarms and missed precipitation. They are also probably more important in the context of applications, which tend to focus on medium-to-heavy rainfall in which hits are prevalent. Prioritizing "hit-relevant" performance such as bias and correlation, we have elected to focus on 2BCMB for the remainder of this study. This issue deserves further attention, however, as the relative performance of 2ADPR and 2BCMB can be expected to vary geographically, seasonally, and with precipitation microphysics [73], [74].

In addition to the absolute estimation performance, another key consideration is the need for approximate (if not strict) independence from the SMP product being evaluated, if DPR-derived precipitation products are to be used as alternative

TABLE I

Contingency Tables of 2ADPR and 2BCMB, Benchmarking Against the Ground Reference GV-MRMS

	$P_{GV\text{-}MRMS} \ge$ 0.1 mm·h^{-1}	$P_{GV\text{-}MRMS} <$ 0.1 mm·h^{-1}
$P_{2ADPR} \ge 0.1$ mm·h^{-1}	931,165 (4.4%)	52,187 (0.2%)
$P_{2ADPR} < 0.1$ mm·h^{-1}	44,407 (0.2%)	19,958,348 (95.1%)
$P_{2BCMB} \ge 0.1$ mm·h^{-1}	833,803 (4.0%)	324,859 (1.5%)
$P_{2BCMB} < 0.1$ mm·h^{-1}	141,769 (0.7%)	19,685,676 (93.8%)

Notes: For each pair of estimates, hits (top left), false alarms (top right), misses (bottom left), and correct non-detects (bottom right) are shown. The total paired data sample size over CONUS is 20,986,107.

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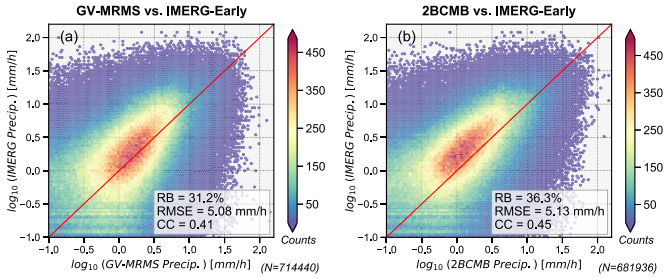

Fig. 3. Density scatterplots of coincident precipitation estimates from GV-MRMS and 2BCMB versus IMERG-E. Only precipitation estimates greater than 0.1 mm·h^{-1} are considered, including all the data samples over the CONUS during the study period.

references. Khan et al. [43] and You et al. [42] argued that the independence need only be approximately met, as numerous processing steps and assumptions stand between the DPR/GMI observations and their manifestation within IMERG (as highlighted in Section II-A). Nonetheless, we examined this by comparing the accuracy of IMERG-E (for hits only) relative to both 2BCMB and GV-MRMS (see Fig. 3). The RBs between IMERG-E and the two reference datasets are similar (31% in the case of GV-MRMS, and 36% for 2BCMB; while visual inspection shows different conditional bias features). RMSEs are very similar (5.08 versus 5.13 mm·h^{-1}), while Pearson CC with 2BCMB is slightly higher (0.45) than with GV-MRMS (0.41). Contingency tables corresponding to Fig. 3 are shown in Supplemental Table S-I and reveal similar detection skills of IMERG-E relative to 2BCMB and GV-MRMS. Taken together, the results suggest that there is indeed approximate independence between IMERG-E and 2BCMB, confirming the latter's potential to evaluate the former. This issue is discussed further in Section V-A.

Climatological CSGDs Fitted to GV-MRMS and 2BCMB share similar spatial patterns of μ and POP (Supplemental Fig. S2a–d; σ and δ were also investigated but are not shown). The 2BCMB tends to slightly underestimate μ and POP relative to GV-MRMS, likely reflecting its imperfect detection and quantification. Fitted CDFs for climatological CSGDs are illustrated for a 1° × 1° box in the state of Tennessee in the Southeastern CONUS and a 1° × 1° box in New Mexico in the Southwest (Fig. S2e–f), which are randomly selected to represent the locations characterized by different climates. Although these CSGDs closely match the empirical CDFs over the more humid Southeastern box, 2BCMB exhibits a higher probability of zero precipitation and relatively large differences from GV-MRMS for light precipitation rates less than 1 mm·h^{-1} (Fig. S2e). In the drier Southwest, a small negative bias in estimated POP is evident (Fig. S2f), consistent with previous studies and related to the CRPS minimization scheme [72].

B. CSGD Error Model Visual Inspection and Deterministic Performance

Linear and nonlinear versions of the CSGD error models trained by GV-MRMS and 2BCMB are further compared over the 1° × 1° boxes in the Southeast and Southwest CONUS (Fig. 4; see Fig. S3 for identical results plotted on

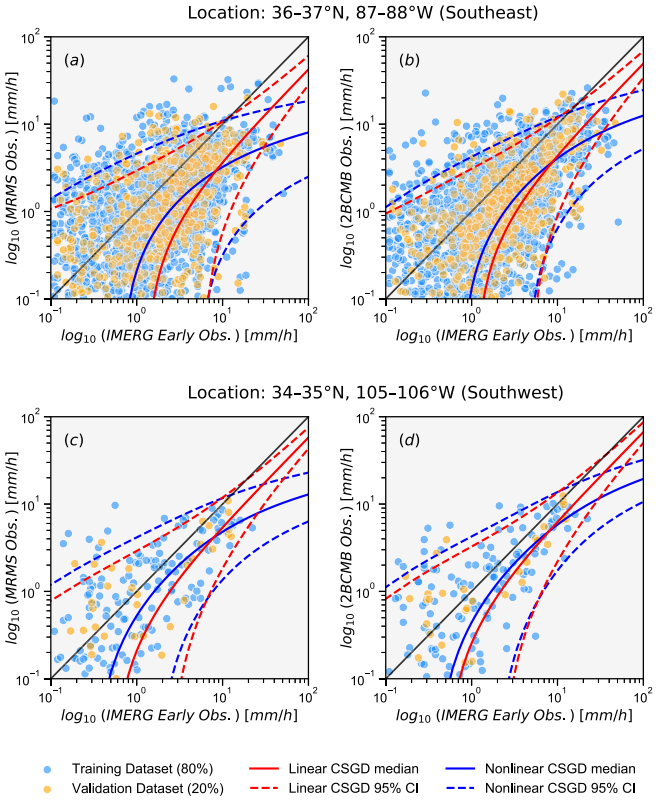

Fig. 4. 95% CI and median of linear (red lines) and nonlinear (blue lines) CSGD error models for: (a) and (b) Southeast 1° × 1° box and (c) and (d) Southwest 1° × 1° box, trained and compared against (Left) GV-MRMS and (Right) 2BCMB. See Fig. S2 for identical results, but plotted on linear scales. Blue dots are training dataset, and orange dots are validation dataset.

linear rather than log–log scales). For these selected boxes and other locations in the CONUS, IMERG-E is prone to overestimate precipitation at half-an-hour scale, particularly at higher precipitation rates. This is consistent with previous regional studies [75], [76], and CONUS-wide analysis [53].

The nonlinear CSGD error models in Fig. 4 perform better than linear models for capturing the conditional bias that is evident at high precipitation rates (e.g., >10 mm·h^{-1}). Visual inspection suggests that the 2BCMB-trained models have similar features to the GV-MRMS-trained models, though the nonlinear versions show slightly weaker systematic bias and a wider uncertainty bound (shown as the 95% confidence interval (CI) in Fig. 4). Both GV-MRMS- and 2BCMB-trained CSGD models provide reasonable fits to the validation dataset, showing the robustness of this uncertainty quantification framework. The sample size of coincident data for the more arid southwestern box [Fig. 4(c) and (d)] is limited due to lower POP, while IMERG-E systematic bias is somewhat lower than in the Southeast [Fig. 4(a) and (b)]. These results highlight the relative flexibility of this CSGD-based uncertainty quantification method under very different climatic conditions.

The central tendency (i.e., means or medians) predicted by the CSGD error model represents the reducible IMERG-E error (i.e., bias; see [37]). We compare the CSGD medians predicted by the GV-MRMS- and 2BCMB-trained models over CONUS against GV-MRMS observations using MAE

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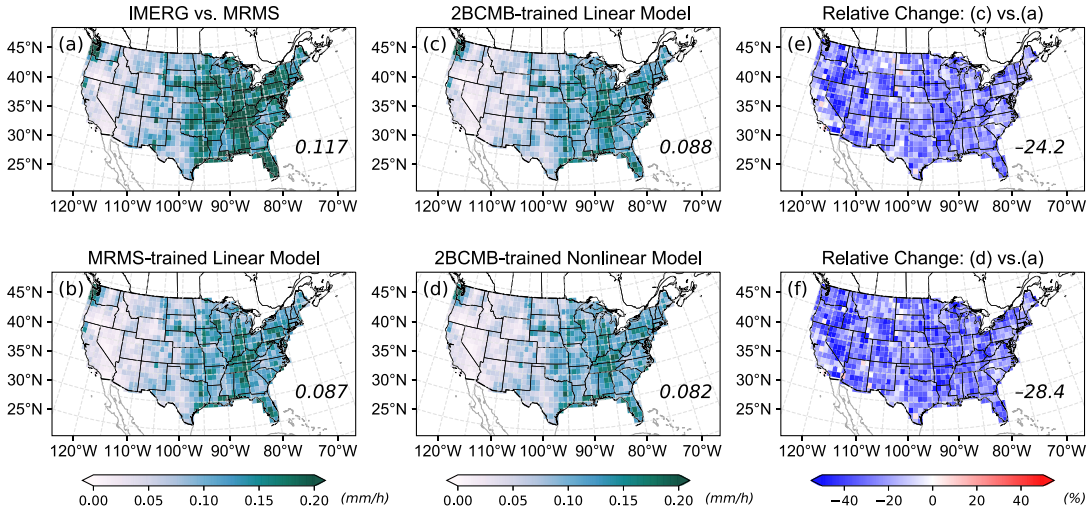

Fig. 5. MAE calculated relative to GV-MRMS for: (a) IMERG-E, and the median predicted by: (b) GV-MRMS-trained linear model; (c) 2BCMB-trained linear model; and (d) 2BCMB-trained nonlinear model. Relative percentage change of MAE relative to IMERG-E results in (a): (e) 2BCMB-trained linear model and (f) 2BCMB-trained nonlinear model. All the results are calculated by validation data samples, and inset values are the means of all 1° × 1° boxes in the CONUS.

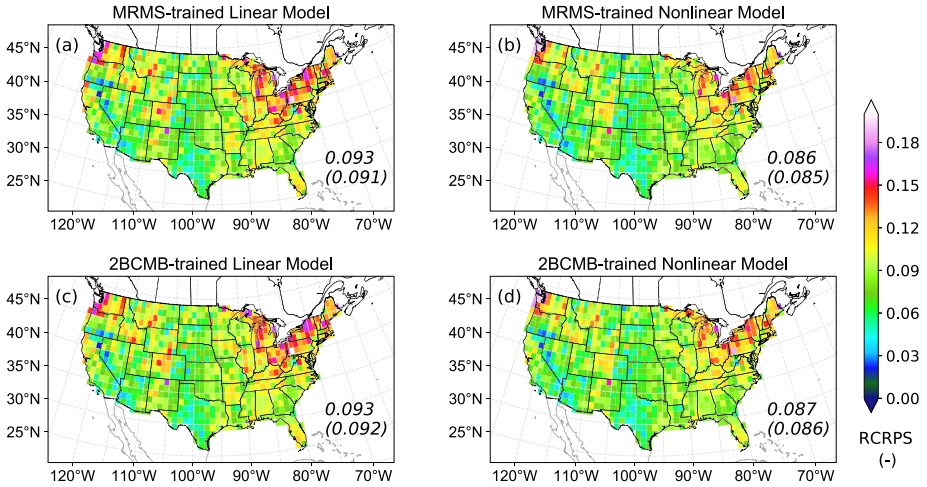

Fig. 6. Comparison of mean RCRPS maps for: (a) GV-MRMS-trained linear model; (b) GV-MRMS-trained nonlinear model; (c) 2BCMB-trained linear model; and (d) 2BCMB-trained nonlinear model. Inset values are the means of all 1° × 1° boxes in the CONUS, and the results for models with $GWTD_{TOP}$ are in parentheses.

[Fig. 5(a)–(d)]. The 2BCMB-trained linear (nonlinear) error models can isolate bias in IMERG-E—reducing MAE by 24.2% (28.4%) on average [Fig. 5(e) and (f)]. The GV-MRMS-trained model performs similarly (see Fig. 5(d) for the linear model; nonlinear model not shown). There is no obvious change in MAE (as well as its probabilistic equivalent CRPS; Fig. 6) after incorporating $GWTD_{TOP}$ (inset statistics in Fig. 6) or other MERRA-2 predictors (results not shown), indicating that these variables have limited roles in explaining IMERG-E systematic error.

C. CONUS-Wide Probabilistic Evaluation

Fig. 6 presents the mean RCRPS maps for different error models over CONUS. The results show somewhat lower CONUS-averaged RCRPS for the nonlinear GV-MRMS- and 2BCMB-trained models (0.086 and 0.087, respectively) compared to the linear models (0.093 for both the GV-MRMS- and 2BCMB-based models). This highlights both the potential of 2BCMB as a reference product and confirms the superiority of the nonlinear model. Interestingly, the areas with highest RCRPS—e.g., Northeast CONUS, the Rockies, the Great Lakes states, and the Pacific Northwest—still show relatively high RCRPS values after considering nonlinear bias. This reflects region-dependent uncertainties of IMERG-E, likely associated with relatively high precipitation totals combined with "complicating factors" such as orography, lake-effect snow, and large fractions of annual precipitation falling as snow.

D. Model Complexity and Conditional Performance

We compared CONUS-wide performance improvements for a range of GV-MRMS- and 2BCMB-trained error models with varied model complexities, measured by the percentage

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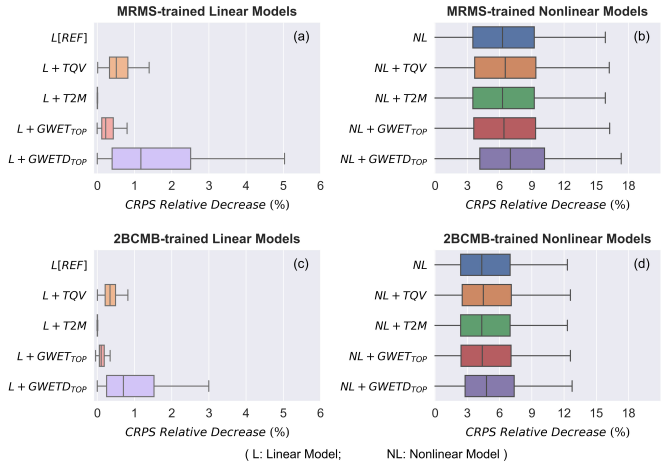

Fig. 7. Boxplots of the percentage decrease in CRPS relative to the linear model with no predictors for: (a) and (b) GV-MRMS-trained error models and (c) and (d) 2BCMB-trained error models with various model complexities and predictors. The results are calculated based on training data samples, including all the 1° × 1° boxes in the CONUS. The covariates TQV, T2M, and $GWET_{TOP}$ represent the total precipitable water vapor, 2-m air temperature, and topmost soil layer's ground wetness index from MERRA-2, respectively, and $GWETD_{TOP}$ is the ground wetness change indicator that is derived from $GWET_{TOP}$. Please note the different x-axis scales between (a) and (c) and (b) and (d).

decrease of CRPS relative to the linear model without any covariate (Fig. 7). Consistent with Fig. 4, nonlinearity is the most critical model feature for constraining uncertainty estimates (i.e., improving CRPS). Moreover, the improvement gained via the nonlinear formulation tends to be larger for the GV-MRMS-trained model than for the 2BCMB-trained model (a mean of 6% versus 4%, respectively), consistent with the larger conditional bias "detected" by GV-MRMS as shown in Figs. 4 and S2. Fig. 7 also indicates that the most informative covariate we evaluated is $GWTD_{TOP}$, which can provide modest improvements (generally 0.5%–1.5% reduction in CRPS, ranging as high as 5% in some locations) to both the linear and nonlinear models. On average, the other covariates—TQV, T2M, and $GWTD_{TOP}$—provide more limited benefits. Therefore, only $GWTD_{TOP}$ is shown elsewhere in this study.

We further group the validation dataset and corresponding error model predictions into four cases: hits, misses, false alarms, and CNs. Table II summarizes CONUS-wide mean CRPS values of the four categories. Since the number of instances of the groups differ widely, we also show an overall "weighted mean (WM)" CRPS for each error model. In general, the 2BCMB-trained model shows similar performance as GV-MRMS-trained model in characterizing both the overall uncertainty and the errors associated with these four cases. The results highlight that the role of model nonlinearity and $GWTD_{TOP}$ may vary among different cases. The nonlinear model shows improved performance in both hits and CNs cases, but performs worse than the linear model for misses and false alarms. Because the CNs and hits dominate in the coincident samples (accounting for 92.6% and 3.4% of the validation dataset, respectively), the WM CRPS of the nonlinear model outperforms that of the linear version.

At the same time, $GWTD_{TOP}$improves the uncertainty estimates for hits and misses, and the latter particularly.

TABLE II

CONUS-WIDE MEAN CRPS FOR DIFFERENT CASES: HITS (H, 3.4% OF VALIDATION DATA), MISSES (M, 2.0%), FALSE ALARMS (F, 2.0%), AND CNS (92.6%), AND THEIR WM ACCORDING TO THE PERCENTAGE OF DIFFERENT CASES IN VALIDATION DATASET

Cases	Mean CRPS for different error model complexities ($mm \cdot h^{-1}$)			
	GV-MRMS-trained Linear Model		GV-MRMS-trained Nonlinear Model	
	No Covariate	With $GWTD_{TOP}$	No Covariate	With $GWTD_{TOP}$
H	1.484	1.449	1.282	1.275
CN	0.0009	0.0013	0.0003	0.0007
M	0.913	0.889	0.924	0.903
F	0.132	0.131	0.219	0.214
WM	0.072	0.071	0.067	0.066
	2BCMB-trained Linear Model		2BCMB-trained Nonlinear Model	
	No Covariate	With $GWTD_{TOP}$	No Covariate	With $GWTD_{TOP}$
H	1.497	1.470	1.330	1.325
CN	0.0004	0.0007	0.0001	0.0003
M	0.922	0.904	0.931	0.916
F	0.133	0.132	0.190	0.186
WM	0.072	0.071	0.067	0.067

Notes: The four cases are divided according to GV-MRMS and the coincident IMERG-E observations (using the same 0.1 $mm \cdot h^{-1}$ rain/no-rain threshold). The percentages of each case are similar to those in Table S1, but only the validation dataset is considered here.

$GWTD_{TOP}$ can indicate precipitation occurrence using changes in land surface wetness conditions, and thus, it helps in quantifying the "missed" precipitation that is difficult to address in most SMP products, even with gauge corrections [29], [77]. However, the inclusion of $GWTD_{TOP}$ also increases CRPS in cases of CNs.

To better characterize the magnitude-dependent performance of the uncertainty estimates for hits case, we consider 0.1-, 1-, 5-, 10-, and 25-$mm \cdot h^{-1}$ thresholds, which roughly correspond to 0.97, 0.99, 0.995, and 0.998 quantiles of IMERG-E climatology. Fig. 8 shows kernel-based density functions of CRPS for validation IMERG-E data between these thresholds. For IMERG-E detected precipitation events, the nonlinear error model outperforms the linear model, as the distributions of CRPS of the former are more concentrated toward zero and thus have a smaller mean. For larger IMERG-E estimates, differences become more apparent and the nonlinear models yield smaller CRPS scores than the linear models [Fig. 8(c) and (d)]. This is unsurprising since IMERG-E shows substantial nonlinear conditional bias at high precipitation rates (e.g., Fig. 4). This analysis also conveys that the 2BCMB-trained models perform similar to the GV-MRMS-trained model at different intensities, showing a slight degradation for very heavy precipitation rates (>25 $mm \cdot h^{-1}$). The benefits of adding $GWTD_{TOP}$ are not visually obvious in these CRPS-based analyses; however, it does offer very modest benefits,

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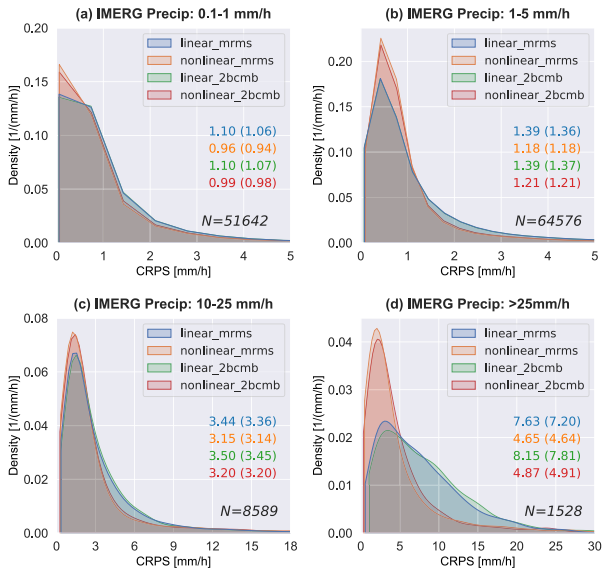

Fig. 8. Distributions of CRPS for GV-MRMS- and 2BCMB-trained error models for different intervals of IMERG-E precipitation rates (validation data samples of "hits" only; N is the sample size). Inset values are the mean CRPS [mm·h^{-1}] for the different models. Values inside (outside) the parentheses include (do not include) GWTD$_{TOP}$ as a predictor. Please note the different x-axis scales between the four subpanels: (a) and (b) CRPS: 0-5 mm/h; (c) CRPS: 0-18 mm/h; and (d) CRPS: 0-30 mm/h.

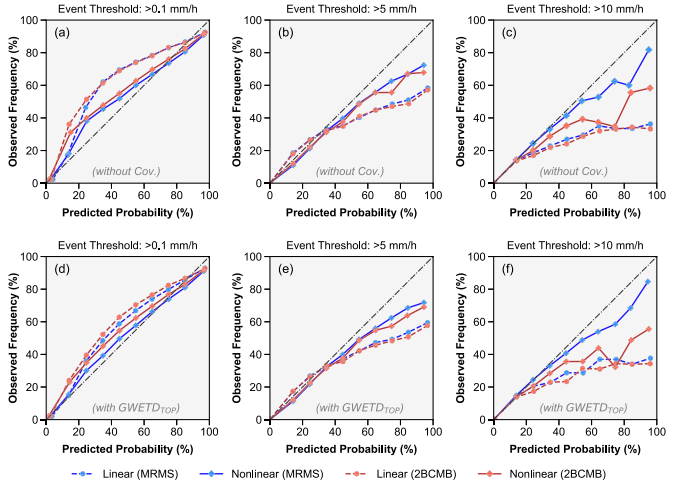

Fig. 9. Reliability diagrams for GV-MRMS- and 2BCMB-trained error models with various precipitation event thresholds. The observed frequency is calculated based on the validation GV-MRMS data samples. The top panels are for models without covariates, and bottom panels are for models with GWTD$_{TOP}$.

particularly in the linear models at higher precipitation rates (inset statistics in Fig. 8).

Reliability diagrams for all the error models with thresholds of 0.1, 5, and 10 mm·h^{-1} are compared in Fig. 9. The results clearly show that the nonlinear model always offers more reliable estimates than the linear model (as it always falls closer to the 1:1 line). The 2BCMB-trained nonlinear model presents similar skill to the GV-MRMS-trained model, except for events larger than 10 mm·h^{-1}. The model reliability varies at different event thresholds. For precipitation occurrence (i.e., 0.1 mm·h^{-1}), all error models tend to be consistently above the diagonal, particularly at low-medium forecast probability categories. This feature indicates that the observed frequency in each category exceeds the model estimated frequency (i.e., a "dry bias"; see [68]). This is similar to the findings of an earlier study on CSGD models [72] that relates this dry bias to the underestimation of POP by climatological CSGDs (which can be seen in Fig. S2f). This dry bias is reduced by incorporating GWTD$_{TOP}$, which is unsurprising since it serves as an "indicator" of precipitation occurrence. For heavy events, on the other hand, all error models fall below the diagonal at high forecast probability categories, indicating the model tends to overestimate occurrence frequency. As discussed in [78], however, the uncertainty in observed frequency increases at higher precipitation rates due to limited sample sizes.

V. Discussion

A. DPR-Derived as Reference Datasets

Our findings point out the promise of DPR-derived precipitation products—and particularly 2BCMB—to serve as a reference in the proposed global IMERG uncertainty estimation framework. However, it must be recognized that while 2BCMB generally outperforms 2ADPR over CONUS (Fig. 2; though not in terms of precipitation detection; see Table I), this conclusion may not hold in other places around the globe. The rejection of 2ADPR as a reference here is not inconsistent with Khan et al. [43], who showed that conditional biases in IMERG and 2ADPR are similar.

Despite the similar error modeling results between 2BCMB- and GV-MRMS-trained error models, the two references differ in important ways: 2BCMB estimates instantaneous precipitation rate, while GV-MRMS offers precipitation estimates aggregated into 30-min intervals. This scale mismatch will inevitably introduce additional uncertainties, which are likely to manifest in the form of random errors both in precipitation occurrence and magnitude. This seems unlikely to influence systematic errors. It should be noted that 2BCMB underestimates high precipitation rates (Fig. 2), probably due to the attenuation of radar signals and estimation of the radar profile between the DPR's lowest bin and the surface. This likely explains the somewhat degraded performance of the 2BCMB-based error models at high precipitation rates, relative to the performance at lower intensities (Fig. 8).

As mentioned in Sections II-A and IV-A, an objection can be raised to the use of DPR-derived precipitation datasets as reference for IMERG or other GPM precipitation data products, due to the (indirect) inclusion of DPR (and GMI, in the case of 2BCMB) in those products and thus the potential for lack of independence between them and the DPR-based reference. The results in Section IV-A and the arguments of Khan et al. [43] and You et al. [42] suggest that this objection should not be overly concerning. The most direct contribution from DPR and GMI combined data in IMERG algorithm is the 45-day probability matching intercalibration of PMW-only precipitation retrievals, before the morphing and PMW-IR merging procedures that derive the ultimate gridded IMERG estimates [44]. The actual difference in IMERG estimates

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before and after this intercalibration is found to be typically less than 10% [78], [79]. In addition, the combined data are also used to build the a priori databases in GPROF, which then provides the PMW-only precipitation retrievals in IMERG. However, it should be noted that this impact on potential dependence is quite limited since only one-year (September 2014–August 2015) combined data are utilized in constructing that database [80]. In other words, there is not a real-time stream of DPR measurements—and associated random errors—feeding directly into IMERG. Nonetheless, recent work by Kirstetter et al. [20] shows that systematic biases such as those associated with precipitation typology display similar features across Level-2 and Level-3 GPM products (i.e., QPE from DPR, GPROF-GMI, and IMERG), suggesting that uncertainty can propagate from DPR-based precipitation estimates into the SMP product.

B. CSGD-Based Error Model

The evaluation demonstrates that the CSGD error model can reasonably characterize IMERG uncertainty over the CONUS. Consistent with previous studies [37], [81], model fitting was relatively robust to small sample sizes (Fig. 5). This property is crucial if the framework is to be expanded to the entire globe, as GPM DPR sampling frequency decreases at lower latitudes (see Fig. 1(d) and [42, Fig. 1]). It is also worth mentioning that the robustness of fitting was achieved after the spatial pooling of neighboring grid cells. For this reason, the CSGD error models in this study were trained using a "baseline climatology" that was locally constructed within $1^\circ \times 1^\circ$ spatial windows without considering seasonal variation. However, this ad hoc window size and seasonality setting cannot guarantee an adequate estimation of the climatology everywhere around the globe. These two issues need to be further examined in future applications though the latter was found to offer no major advantage in previous CONUS-wide error modeling studies [37], [40].

The inclusion of additional predictors into the model provides a way of improving uncertainty estimates. As demonstrated in this study, incorporating $\mathrm{GWTD_{TOP}}$ derived from the MERRA-2 reanalysis product modestly improved the model's performance in characterizing precipitation occurrence (see Table II and Fig. 9). Expectation of greater gains using MERRA-2 is unwarranted given its coarse temporal and particularly spatial resolution. It is not clear that other satellite-based products could be much help here—consider, for example, that remotely sensed soil moisture data likely lack the combination of global coverage and short latency needed to inform near-real-time IMERG uncertainty estimates. High-resolution global-scale numerical weather forecasts such as those from NASA's Goddard Earth Observing System Forward Processing products (GEOS-FP) [82], on the other hand, seem to offer potential as they are available on a consistent global basis in real time. Another direction which is being explored in separate work is using simple metrics derived from the IMERG precipitation fields themselves as predictors in the error model.

Finally, while we explored several deterministic and probabilistic performance measures in this study, it is doubtful that we have fully explored all relevant aspects of model skill. Future global validation efforts will continue exploring this topic using more candidate evaluation metrics (see [17], [67]) and in varied environmental settings.

C. IMERG Uncertainty Beyond the DPR Swath

This study's central premise—that GPM DPR-derived precipitation products can serve as an alternative reference for estimating IMERG uncertainty—carries a key limitation: since spatial and temporal coincidence between IMERG and DPR is needed to train the error model, this can only occur within the DPR swath. Of course, DPR and GMI are co-located on the GPM-CO. The result is that the uncertainty estimates presented here only reflect the relationships between DPR and GMI-based IMERG retrievals (i.e., GPROF-GMI). Because GMI is one of the most accurate PMW precipitation radiometers currently in space [73], our analyses probably provide the "best scenario" (i.e., lowest uncertainty); the real uncertainty associated with IMERG estimates that are dominated by other PMW or IR sensors or morphing is likely greater (see [8], [19]) than our error models would predict. It should be noted in particular that the current IMERG algorithm abandons PMW (including GMI) retrievals in places with snow/icy surfaces and switches to the PMW-calibrated IR estimates [44]. This likely explains the relatively high uncertainty (identified by all error models) over the Pacific Northwest, the Rockies, the Great Lakes states, and the Northeast CONUS (Fig. 6), where large fractions of precipitation fall as snow.

To overcome this limitation, more work is needed to investigate how DPR-based uncertainty estimates can be "inflated" to better reflect the properties of other PMW and IR sensors as well as IMERG's morphing scheme. Following the same method as in this study, it will be straightforward to separately train an error model for IR-influenced IMERG estimates, as the PMW-calibrated IR precipitation fields are contained in the native IMERG data files. As for other GPM constellation PMW sensor-specific uncertainty estimates, the work of You et al. [42]—who compared DPR-derived precipitation estimates against those of neighboring PMW sensors (when they are less than 5 min apart and 5 km away) within GPROF—provides a possible blueprint for addressing this. In addition, the two-step adjustment described in Section II-A—that climatologically calibrates all the other PMW sensors to GMI and then calibrates GMI to 2BCMB on a short time interval (e.g., 45 days)—may also be worth exploring.

VI. Conclusion

This study proposes a prototype uncertainty quantification framework for global multisensor satellite precipitation products, in which GPM DPR-derived precipitation datasets are used in place of ground-based measurements. Though we focus our study on the IMERG-Early run at its native 30-min, 0.1° resolution over CONUS, the quasi-global availability of DPR measurements allows this framework to be applied across the globe to any satellite precipitation dataset whose spatiotemporal resolution is similar, such as CMORPH and

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PERSIANN. Uncertainty is modeled using a flexible and parsimonious three-parameter CSGD error model which can characterize the POP as well as intensity-dependent systematic bias and the potential range of random error. Uncertainty estimates from the CSGD error model trained on the DPR-based reference are compared against high-quality ground observations from the GV-MRMS dataset, as well as uncertainties inferred from GV-MRMS-trained versions of the error model. We believe this is the first study to generate IMERG uncertainty estimates using this globally applicable approach.

Our CONUS-wide assessment suggests that the combined (DPR and GMI) product 2BCMB outperforms 2ADPR (which uses DPR exclusively) in terms of precipitation intensity statistics. The 2ADPR has somewhat better detection properties, however. Due to substantial intensity-dependent error behaviors in IMERG, the rejection of 2ADPR as a reference by Khan et al. [43], and the supposition that better uncertainty estimates during precipitating periods would be preferable to better estimates of detection uncertainty, we focused our error modeling analysis on the 2BCMB product.

Multiple CSGD-based IMERG error models of varying complexity were trained using both GV-MRMS and 2BCMB. We found that the precipitation climatology characterized by GV-MRMS- and 2BCMB-trained CSGD models yield similar properties and comparable performance throughout CONUS though the 2BCMB-trained model has slightly lower mean and POP values, likely attributable to its imperfect detection skill. The 2BCMB-trained model performance also suffers at high precipitation rates compared with GV-MRMS-trained models. Evaluation using CRPS indicates that IMERG uncertainty, for both models, is relatively high at the Northeast CONUS, the Rockies, the Great Lakes states, and the Pacific Northwest, where a number of "complicating factors" such as orography, lake-effect snow, as well as high fractions of annual precipitation falling as snow may complicate IMERG errors.

Relatively weak error model performance can be ameliorated by incorporating additional predictors to further constrain uncertainty estimates. We illustrate this by incorporating predictors from NASA's MERRA-2 reanalyses, including a derived variable representing positive temporal deviations in near-surface soil moisture that can be associated with precipitation occurrence and that improves uncertainty estimates, albeit modestly. Higher resolution numerical weather predictions—particularly short-range high-resolution global forecasts—could be potentially useful for informing uncertainty estimates for near-real-time versions of IMERG. In addition, variables derived from IMERG's ancillary data or the spatial structure of the IMERG fields themselves offer further promise to constrain IMERG uncertainty.

Despite the promising performance of this uncertainty framework over CONUS, its flexibility and robustness in other parts of the globe remain untested—though the validation is challenging at those places due to the lack of high-quality ground observations. The accuracy of IMERG as well as the relative performance of the 2ADPR and 2BCMB products is influenced by climate, land surface conditions, and atmospheric and precipitation properties. DPR's sampling frequency also varies with latitude due to its inclined orbit. Future work will focus on validating uncertainty estimates in other locations and conditions.

Notwithstanding these remaining challenges, the IMERG uncertainty estimates provided by this general error modeling framework can benefit end-user applications [5]. As illustrated in a recent study, the incorporation of CSGD-based satellite precipitation uncertainty can improve regional landslide hazard nowcasting, even when a CSGD error model is trained by very limited data [82]. It should be noted that the maximum benefits of the IMERG error modeling framework only only be achieved when regional and global environmental modeling systems and workflows are adapted to "ingest" such uncertainty estimates. Proof-of-concept efforts in that direction are equally important as further validation of our uncertainty quantification approach.

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